

Hanlei Zeng

XR / HCI Researcher · Interaction Design for Digital Cultural Heritage · Embodied Interaction & Spatial Experience Research

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Research Interests

My research focuses on **embodied interaction design at the intersection of HCI, XR, and digital cultural heritage**, exploring how **vision-based markerless gesture interaction, spatial computing, and natural user interfaces** can enhance interaction between people and digital environments. Through a **Research through Design** methodology, I develop interactive prototypes and study their applications in **digital heritage exhibitions, immersive experiences, and spatial narrative design**. I am also interested in the potential of **interactive lighting as a digital medium**, exploring mappings between light, bodily movement, and cultural content.

Research Experience

Gesture-Driven Embodied Interaction for Digital Heritage

Independent research & prototyping | 2025 – Present

Exploring how vision-based markerless gesture interaction can foster user participation and cultural experience, centered on **HCI, XR, and digital heritage interaction**.

- Developed browser-based 3D heritage interaction prototypes (**JavaScript, MediaPipe, Three.js**) around bronze artifacts and Dunhuang digital heritage, combining 3D models, gesture input, and visual feedback to explore embodied cultural participation.
- Developed mixed-reality spatial interaction prototypes (**Unity, C#, MRTK, HoloLens 2**), building user-needs models with grounded theory to realize spatial visualization and interactive customization. The project originated from the Zhenjiang Municipal Key R&D Program (GY2021014).
- Principal investigator of **JAT251371 (VR-based collaborative lighting design research)**, exploring the combination of XR spatial interaction and interactive lighting design.

Research Methods

Qualitative: Grounded theory, semi-structured interviews, participatory design, contextual inquiry **Quantitative:** Experimental design, SEM, DEMATEL, questionnaire analysis, usability evaluation **Design research:** Research through Design, interactive prototype development, iterative design

Research Projects

- **PI:** JAT251371, VR-based collaborative lighting design research.
- **Participant:** GY2021014, research on a personalized customization system for modified vehicles based on mixed reality (Zhenjiang Municipal Key R&D Program).
- **Participant:** 2020 university-level innovation fund project, cross-regional medical system design based on 5G and VR.

Research Output

Papers

Sha, C., Fang, Y., **Zeng, H.**, Han, R., & Donkor, E. K. N. (2026). *A mixed reality-based personalization system for recreational vehicle interior customization*. **Tehnički vjesnik – Technical Gazette**.

Sha, C., **Zeng, H.**, Wu, B., et al. (2022). *Research on the application of mixed reality technology in the field of art design*. **SHS Web of Conferences**, 145.

Zeng, H. (2023). *Design research on a mixed reality-based customization system for RV interiors.* Master's Thesis, Jiangsu University.

Patents & Software Copyright

A method and system for customizing motorhome interiors based on mixed reality technology (CN114782660A).
MR RV Interior Personalization System (2022SR0889245).

Work Experience

- 1. Quanzhou Huaguan College,** Director of the Digital Intangible Cultural Heritage Design and Production Teaching & Research Office | 2025.11 – Present Responsible for teaching and research development in digital intangible cultural heritage; teach 3D character design, digital sculpting and painting, and digital creative product design; supervise student digital content production and visual expression.
- 2. OSRAM / LEDVANCE,** Hardware Product Manager | 2024.10 – 2025.10 Participated in lighting product R&D and commercialization, owning the product process from concept design to mass production, and optimizing products through user research, light-distribution analysis, and spatial experience design.
- 3. LEEDARSON,** Hardware Product Manager | 2023.07 – 2024.07 Conducted user research for the European and American smart lighting markets, extracting user needs with data analysis and the DEMATEL method, and driving product development.
- 4. TCL,** IoT Product Manager Intern | 2022.05 – 2022.12 Conducted smart-home scenario research; completed user interviews, interaction flow design, and product prototyping.

Education

Jiangsu University | M.F.A., Digital Art Design | 2020 – 2023 National postgraduate entrance examination: 402/500, ranked 1st in both the written examination and the interview.

College of Science and Technology, Hunan University of Technology | B.A., Product Design | 2016 – 2020

Technical Skills

XR / Interaction : Unity · C# · HoloLens 2 · MRTK · JavaScript · MediaPipe · Three.js
3D / Digital Media : ZBrush · Maya · Rhino · Cinema 4D · Blender · KeyShot
Design Tools : Figma · Photoshop · Illustrator
Research Methods : Grounded Theory · SEM · DEMATEL · User Study · Usability Evaluation
Lighting & AI Workflow : DIALux · Generative AI · Python Workflow Automation

Awards

Bauhaus International Design Competition (Silver Award / Excellence Award) · Milan Design Week (Second Prize) · Guoqing Cup (Second Prize) · Huacan Award (Third Prize) · National College Digital Art Design Competition (Third Prize) · Rural Museum Contemporary Narrative Creative Design Competition (First Prize, etc.)